

When an entity dies, the combat tracker only records the killing blow

MC-121048 Combat

Status	Resolved / Fixed
Affected	Minecraft 1.12.2, Minecraft 1.13.2, 1.15.2, 20w09a, 20w10a, 20w11a (+47 more)
Fixed in	23w03a
Reported	2017-10-08

STEPS TO REPRODUCE

1. Launch Minecraft and create a new Survival world with default settings.
2. Encounter a living entity and allow it to damage you without killing you.
3. Move to a height from which a fall will be fatal.
4. Fall from that height and die.
5. Observe that the death message does not identify the earlier attacking entity in the "doomed to fall" death message.

OBSERVED BEHAVIOUR

Actual behavior

When a living entity takes lethal damage, `LivingEntity#actuallyHurt` reduces its health to zero before calling `CombatTracker#recordDamage`. As a result, `CombatTracker#recheckStatus` sees that the entity is no longer alive and clears the existing combat history before recording the lethal damage entry.

Consequently, the combat tracker contains only the killing blow. When the entity subsequently dies and `getKillCredit` queries the tracker, earlier attacks are no longer available to identify the responsible attacker. This prevents combat-history-dependent death messages, such as the "was doomed to fall by ..." message, from being selected when appropriate; the entity instead receives a generic death message based only on the final damage source.

EXPECTED BEHAVIOUR

When an entity dies, its combat tracker should retain the complete relevant damage history, including attacks made before the killing blow. The game should use this history to correctly identify the responsible attacker and display the appropriate death message, including "doomed to fall" when applicable, rather than retaining only the final damage entry.

ENVIRONMENT

Minecraft Java Edition Versions: 1.12.2, 1.13.2, 1.15.2, 20w09a, 20w10a, 20w11a, 20w12a, 20w14a, 20w15a, 20w16a, 20w17a, 20w21a, 1.16 Pre-release 2, 1.16 Pre-release 3, 1.16 Pre-release 4, 1.16 Pre-release 7, 1.16 Pre-release 8, 1.16 Release Candidate 1, 1.16, 1.16.1, 20w27a, 20w30a, 1.16.2, 1.16.3, 20w51a, 21w14a, 21w18a, 1.17 Release Candidate 1, 1.17, 1.17.1 Pre-release 1, 1.17.1, 21w37a, 21w42a, 21w43a, 1.18 Pre-release 4, 1.18.1, 1.18.2, 22w12a, 22w14a, 22w15a, 22w17a, 1.19 Pre-release 2, 1.19, 1.19.1 Pre-release 4, 1.19.1 Pre-release 6, 1.19.1, 1.19.2, 22w42a, 22w43a, 22w44a, 22w46a, 1.19.3 Pre-release 1, 1.19.3 (includes snapshots)

ORIGINAL DESCRIPTION

Code analysis below uses 1.19's code, deobfuscated manually using the official obfuscation mappings for 1.19.

Living entities have a `CombatTracker`, which tracks the details of recent hits. When such an entity is damaged, the combat tracker's `recordDamage` method is called to record it.

`recordDamage` first calls `recheckStatus`, which resets the tracked data if certain conditions are met. It then adds the data for the damage currently being taken.

The issue here is that, when `recordDamage` is called for the killing blow, `recheckStatus` will clear the tracker due to the condition `!this.mob.isAlive()` being met. This is because `isAlive` checks that the entity's health is above 0.

Looking at the method `LivingEntity#actuallyHurt` shows the problem:

```

protected void actuallyHurt(DamageSource damageSrc, float amount) {
if (!this.isInvulnerableTo(damageSrc)) {
amount = this.getDamageAfterArmorAbsorb(damageSrc, amount);
amount = this.getDamageAfterMagicAbsorb(damageSrc, amount);
float f2 = amount;
amount = Math.max(amount - this.getAbsorptionAmount(), 0.0F);
this.setAbsorptionAmount(this.getAbsorptionAmount() - f2 - amount);
float f = f2 - amount;
if (f > 0.0F && f < 3.4028235E37F && damageSrc.getEntity() instanceof ServerPlayer) {
((ServerPlayer)damageSrc.getEntity()).awardStat(Stats.DAMAGE_DEALT_ABSORBED, Math.round(f * 10.0F));
}
if (f2 != 0.0F) {
float f1 = this.getHealth();
this.setHealth(f1 - f2);
this.getCombatTracker().recordDamage(damageSrc, f1, f2);
this.setAbsorptionAmount(this.getAbsorptionAmount() - amount);
this.gameEvent(GameEvent.ENTITY_DAMAGE);
}
}
}

```

The health is updated before recordDamage is called, which makes the entity show up as dead and causes the combat data to be cleared.

Then, shortly afterwards, die is called for the just-killed entity. This checks getKillCredit to determine who to credit with the kill.

getKillCredit will first consult CombatTracker#getKiller which contains logic to go through the combat history and pick out the "best" attacker.

However this logic is currently irrelevant, as due to the above issue, the tracker will only contain an entry for the killing blow and nothing else.

The solution here is to change Liv

MANUAL RESULT

Version used	
Reproduced?	YES / NO / PARTIAL
Crash file	
Notes	
Tested by / date	

Live issue: <https://bugs.mojang.com/browse/MC/issues/MC-121048>